

be estimated to the correct order of magnitude.

29	C	An isotope of the parent nuclide will have the same number of protons but different number of neutrons. Hence the <u>atomic number of the isotope is the same</u> as the parent nuclide while its <u>mass number is different</u> after the decays. alpha particle – ${}^4_2\text{He}^{2+}$, beta particle – ${}^0_{-1}\text{e}$, gamma particle – massless, no charge Option C (correct): One alpha decay – mass number decreases by 4, atomic number decreases by 2 Two beta decays – mass number remains the same, atomic number increases by 2 Overall – daughter nuclide <u>mass number decreases</u> by 4, <u>atomic number remains the same</u> Option A (incorrect): The release of a gamma photon does not affect the atomic and mass numbers. Option B (incorrect): One alpha decay – mass number decreases by 4, atomic number decreases by 2 One beta decay – mass number remains the same, atomic number increases by 1 Overall – daughter nuclide mass number decreases by 4, atomic number decreases by 1 Option D (incorrect): Two alpha decays – mass number decreases by 8, atomic number decreases by 4 One beta decay – mass number remains the same, atomic number increases by 1 Overall – daughter nuclide mass number decreases by 8, atomic number decreases by 3
30	B	